



11/6/18

Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India
(Autonomous College Affiliated to University of Mumbai)

End Semester Examination May - 2018

Max. Marks: 100

Class: TE

Course Code: ETC605

Name of the Course: Operating Systems

Duration: 3 Hours

Semester: VI

Branch: EXTC

Instruction:

- (1) All questions are compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Q No.		Max. Marks	CO																		
Q.1 (a)	List various software components of a typical operating system and hence enumerate various functions and services provided by an operating system.	10	CO1																		
Q.1 (b)	What is a semaphore? Write a small section of code for producer process and consumer process with bounded buffer problem using semaphores to eliminate critical section problem.	10	CO2																		
Q.2 (a)	Consider a following set of processes, with length of CPU bursts given in milliseconds as follows: <table><tr><th>Process</th><th>Burst Time</th><th>Arrival Time</th></tr><tr><td>P1</td><td>10</td><td>0</td></tr><tr><td>P2</td><td>29</td><td>0</td></tr><tr><td>P3</td><td>3</td><td>0</td></tr><tr><td>P4</td><td>7</td><td>0</td></tr><tr><td>P5</td><td>12</td><td>0</td></tr></table> 1. Draw the Gantt charts for FCFS, Non Preemptive SJF and RR (Quantum=10) 2. What is the waiting time of each process for each of the above algorithms? 3. What is the turnaround time of each process for each of the above algorithms? 4. Which algorithm results in the minimum average waiting time?	Process	Burst Time	Arrival Time	P1	10	0	P2	29	0	P3	3	0	P4	7	0	P5	12	0	10	CO2
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P5	12	0																			

OR

Q.2 (a)	Consider a following set of processes, with length of CPU bursts given in milliseconds as follows: <table><tr><th>Process</th><th>Burst Time</th><th>Arrival Time</th></tr><tr><td>P1</td><td>14</td><td>0</td></tr><tr><td>P2</td><td>8</td><td>0</td></tr><tr><td>P3</td><td>21</td><td>0</td></tr><tr><td>P4</td><td>10</td><td>0</td></tr></table> 1. Draw the Gantt charts for FCFS, Non Preemptive SJF and RR (Quantum=10) 2. What is the waiting time of each process for each of the above algorithms? 3. What is the turnaround time of each process for each of the above algorithms? 4. Which algorithm results in the minimum average waiting time?	Process	Burst Time	Arrival Time	P1	14	0	P2	8	0	P3	21	0	P4	10	0	10	CO2
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P4	10	0																
Q.2 (b)	With respect to UNIX OS, Draw and discuss the process state transition diagram. OR	10	CO3															
Q.2 (b)	Discuss various security threats in UNIX OS and also enumerate the features included to address these threats.	10	CO3															
Q.3 (a)	Discuss the mechanism of translating the virtual address (logical address) to its respective physical address for a particular operating system. Consider the concept of segmentation and the case study of Linux OS while writing your answer. OR	10	CO3															
Q.3 (a)	With respect to Linux OS, discuss the file management in general and file allocation method used in particular. (inode structure).	10	CO3															
Q.3 (b)	Consider the following page reference string: 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1 How many page faults would occur for the following replacement algorithms, assuming three page frames? Remember that all frames are initially empty, so your first unique pages will cost one page each. 1. FIFO 2. LRU 3. OPTIMAL	10	CO2															
Q.4 (a)	With respect to file systems, discuss (a) File Attributes (b) File Operations (c) Directory Structure (Single level, Two level and Tree level).	10	CO2															

Q.4 (b)	Assume the number of tracks are 200 which are numbered from 0 to 199 on a particular hard disk. Assume the current head position is on the track number 53. Consider a disk queue with requests for I/O to blocks on cylinders as 98,183,37,122,14,124,65,67. Calculate the average seek time in terms of number of cylinders for the four different disk arm scheduling algorithms (a) FCFS (b) SSTF (c) SCAN (d) C-SCAN	10	CO2
Q.5 (a)	Compare Static priority scheduling algorithm vs Dynamic priority algorithms.	10	CO3
Q.5 (b)	Consider a system with three tasks, which we'll call Task 1, Task 2 and Task 3. Assume all are periodic tasks with periods T1, T2 and T3. Each has a deadline that is the beginning of its next cycle. Task 1 has $P1 = 4$ ms, $C1 = 1$ ms, Task 2 has $P2 = 10$ ms, $C2 = 2$ ms and Task 3 has $P3 = 15$ ms, $C3 = 4$ ms. Consider static priority scheduling (RMA), draw the time space diagram and calculate processor utilization, schedulability.	10	CO3